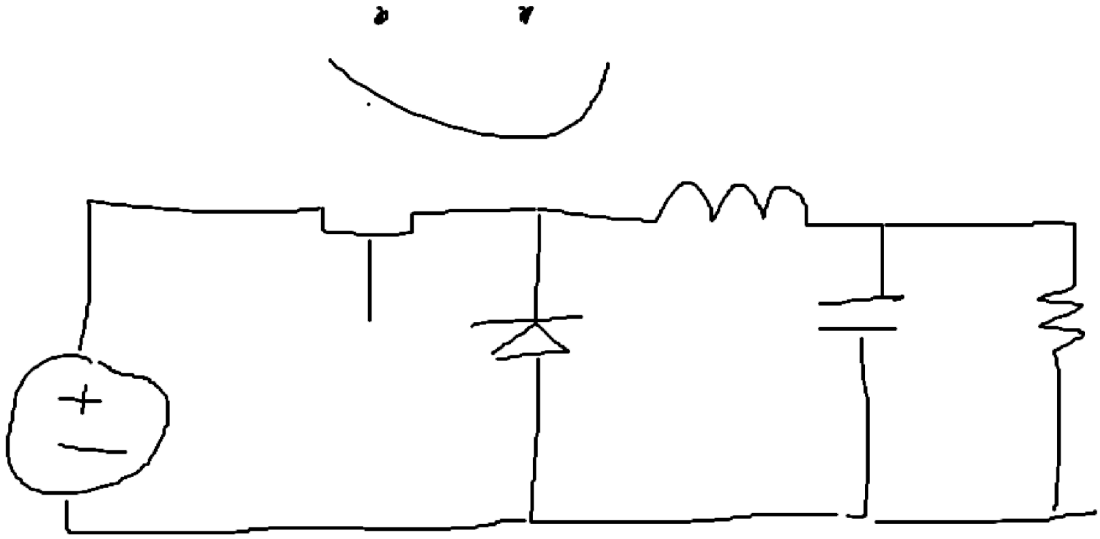
Formula Slug – Buck Converter Schematic Block
Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

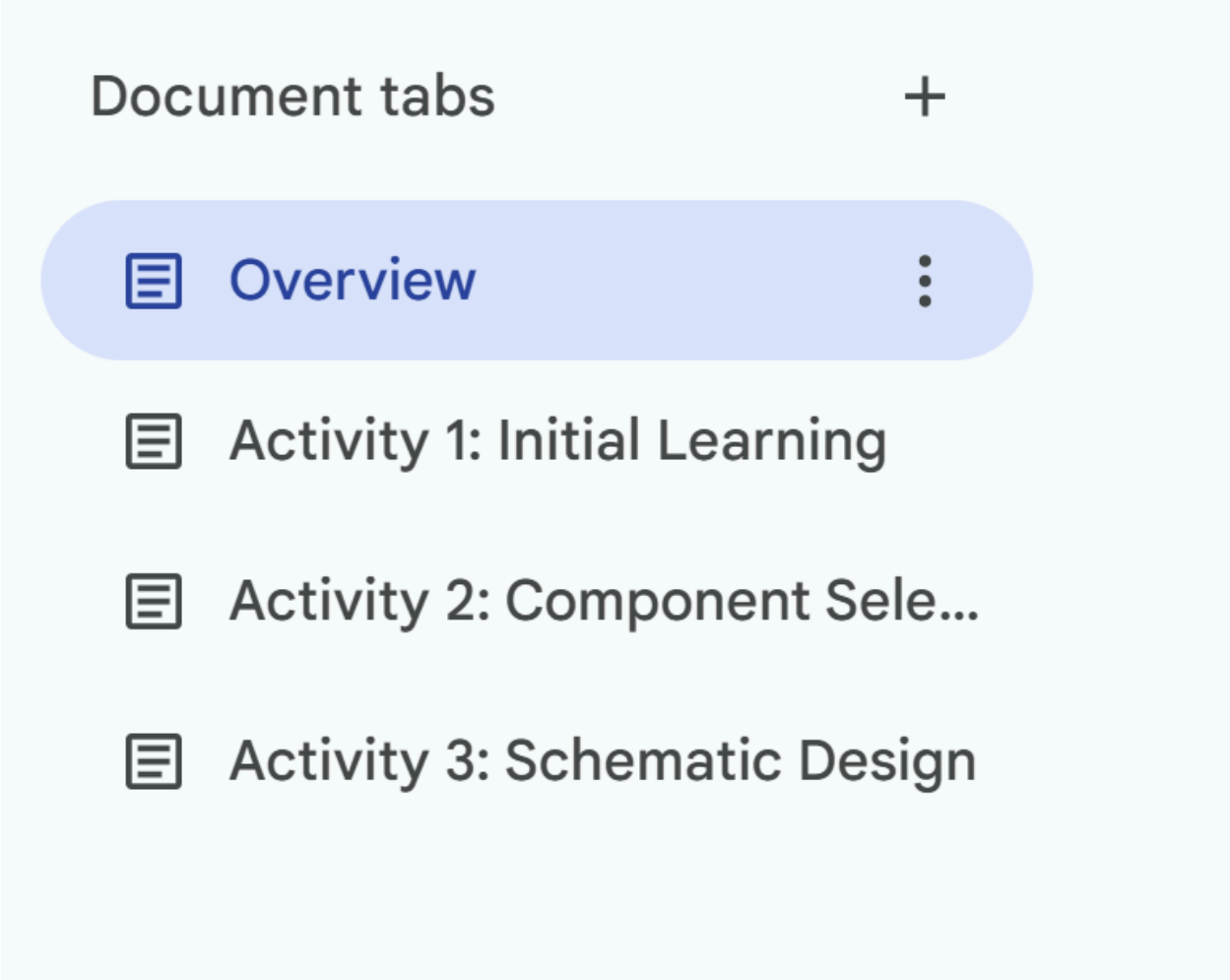
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
   https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
3. Khan Academy, Compound Circuits:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power
4. EEVBlog, Voltage Dividers:
   https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
   https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
   a. Lecture 5: Intro to DC/DC, Part 1
2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

- V drives current through circuit(measured in volts)
- Current I. This is the rate at which electrical charge flows. A for Amps
- Resistance is R. Opposition to the flow of current. Measured in ohms
- Ohm's Law: V=IR
- Constant resistance: Increasing v increases I,constant voltage increasing r decreases I
- Series means one path
- Current is same through every resistor

- Added resistance
- Source voltage divided among the resistors
- Parallel is multiple paths
- Voltage same across every parallel branch
- Current splits between branches
- Equivalent resistance
- Krichoff's voltage law: conservation of energy: voltage rise equals total voltage drops.
- Kirchoff's current law: conservation of charge: total current entering a junction equals total current leaving it
- Series: same current, divided voltage; parallel- same V divided I
- Electric charge: Q measured in coulombs. Electrons carry negative charge
- Electric current, I, rate at which charge passes a point.
- Current of 2A means 2 coulombs of charge pass a point every second
- Conventional current- positive charge directions
- Electrons drift in opposite direction
- V is electrical potential difference
- Volateg is always between 2 points
- R is opposition to current flow: limits flow
- volts/kilo-ohms= milliamps
- 12v /4kΩ = 3mA
- 1kΩ = 1000Ω
- 1MΩ = 1000000Ω
- 1mA = 0.001A
- V voltage across resistor, I is current through resistor
- Fixed resistance: current prop. To voltage 2x V- 2x I,
- Fixed voltage - current is inversely prop. To resistance 2x R - ½ I
- Heating can change resistance
- Ohms not universal for constance resistance model
- I = 1/RV slope = 1/R
- p = material resistivity
- L = conductor length
- A= cross sectional area
- Long conductor: greater resistance
- Larger cross sectional area - lower resistance
- node: electrically connected point or region
- A node is not necessarily a small dot.
- A loop is closed path through circuit
- An open circuit interrupts a conducting path
- A short circuit provides a very low resistance path
- 0 I does not mean 0V, 0V doesn't mean 0I
- Same I pass through: Resistors in series
- Larger series resistance; larger v drop
- Both ends connect to the same two nodes: resistors are parallel
- Electrical power is the rate of energy transfer

- Power measured in watts
- V tells you energy per charge: current tells you energy per second
- P= IV
- Fixed current- P = I^2R
- Fixed voltage- P = V^2R
- resistor - converts electrical energy into thermal energy
- Joule heating
- Energy depends on time
- Kilowatt-hour-unit of energy
- Kirchoff's current law: junction rule: total current entering node equals total current leaving it
- Kirchoff's voltage law: loop rule: signed voltage changes around a complete loop add to zero
- Through a resistor in the assumed current direction: - IR
- Through a resistor against the assumed current direction: + IR
- - actual current flows opposite your arrow.
- - doesn't automatically mean error
- Compound circuit contains both series and parallel connections
- Unloaded means the output connections draws no current
- Series current: I = Vin/R1 + R2
- Output is the drop across R2: V out= IR2
- Vin/R1 + R2 * R2
- Load is whatever you connect to divider's output
- Thevenin equivalent: linear resistive network viewed through 2 terminals
- Vth open circuit output voltage
- vth= vin R2/R1 + R2
- Norton equivalent: same terminal behavior rep by ideal current source IN, parallel resistance RN
- Resistor tolerance: actual resistance can differ from its nominal value
- Voltmeter connects in parallel across 2 points voltage
- Ammeter connects in series with branch current

Activity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



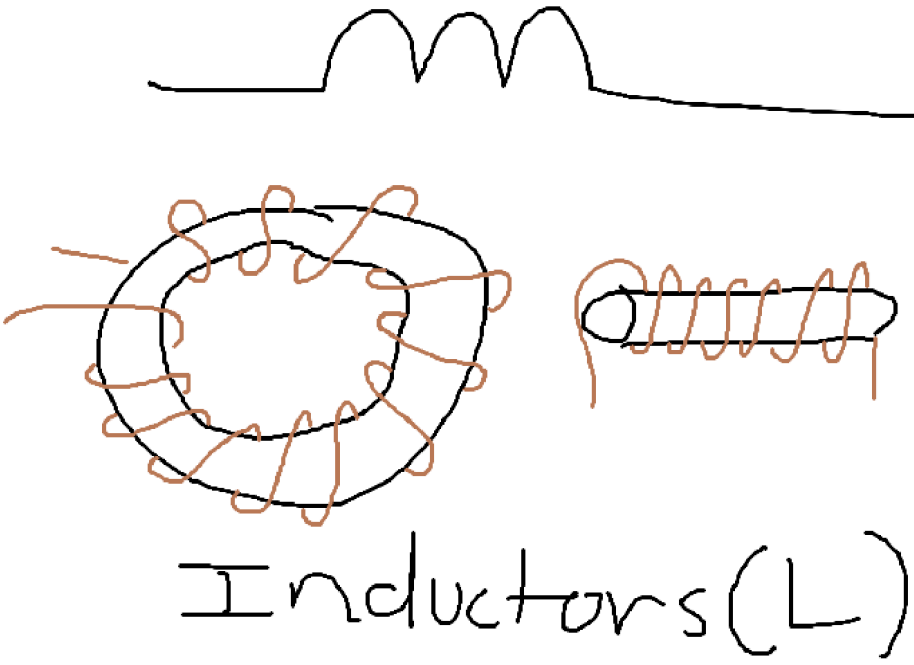
The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

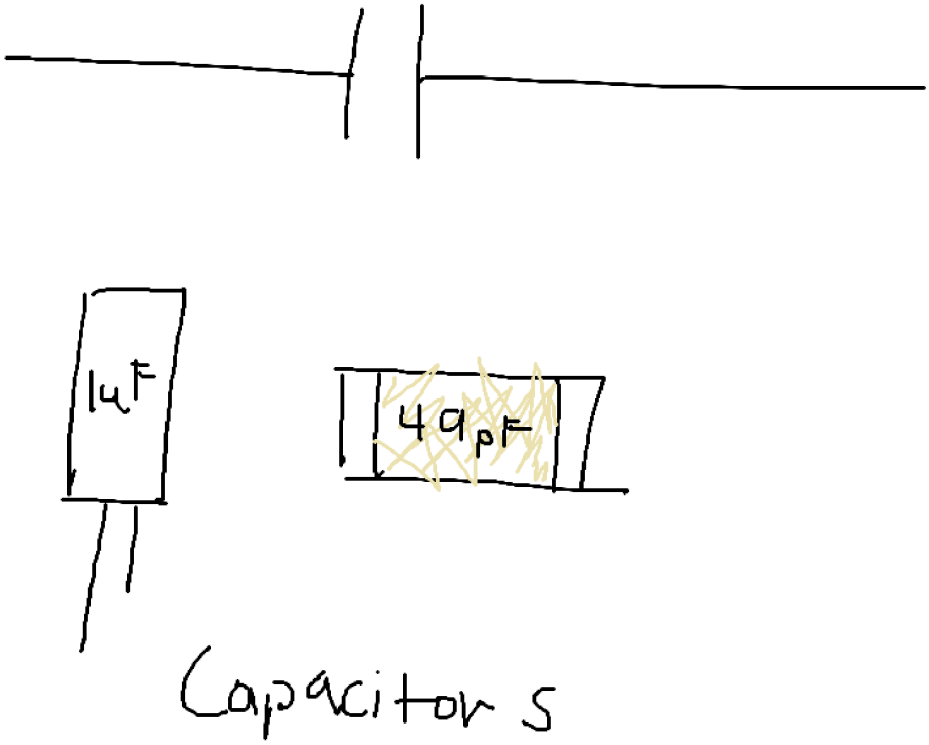
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

Component	Cost	Link	Units	Size
Buck IC	$0.44000 (per unit)	https://www.digikey.com/en/products/detail/texas-instruments/TPS563208DDCR/5825114	1	SOT-23-6 Thin, TSOT-23-6 (6 pin)
Inductor	$1.26000 (per unit) 0.85300 (100 units)	https://www.digikey.com/en/products/detail/codaca/VSEB0630H-3R3MC/16731509	1, 100	6.65mm x 6.45mm Max height: 3.00mm max height
Capacitor	$0.49000 (per unit)	https://www.digikey.com/en/products/detail/murata-electronics/GRM31CR71A226KE15K/4905616	1	3.20mm x 1.60mm 1.80mm max thickness

Total Cost	$2.19(per unit only)			

Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.

Extra Activity: Tractive System Shunt Current and/or Voltage Sense

If you happen to finish the buck converter, or you feel that the buck converter doesn't speak to you, you can also work on this project as well. This one will be in the form of a presentation / research project.

This project is based around adding to our current sensors within our Tractive Battery's container. Currently, we use a hall effect current sensor, which kind of looks like this:



For accuracy and data acquisition reasons, it would be nice to be able to add to this hall effect sensor with a shunt one, where you measure the voltage drop across a specially manufactured piece of metal with a specific resistance and voltage drop across, where you can then determine what current is going through it based on it. However, there are some constraints to this system, ie: isolation, heat dissipation, etc

Our current tractive battery current can go up to ~600A peak. We want to be able to measure up to this and down to the negatives as well in order to track regen current. Required by FSAE 26-27 Rules (EV.8.7.4), we need to be able to *prove* to judges that the Brake System Plausibility Device works through simulating tractive system current going through the current sensor. Find a way to be able to simulate this using a shunt if possible. At the moment, we have wire wrapped around a hall effect sensor multiple times, simulating higher currents with a smaller test current.

We would also like to be able to properly measure the voltage in our tractive system while we drive. We should find a way to take that voltage measurement, isolate it and pass it through to a

microcontroller so that we can have this data for the future.

When completing this research project, continue to write good documentation and make sure to optimize for costs when possible.

Have fun!